

A Retroactive Contact Channel for Behavior Cloning on Low-Cost Arms

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Abstract: Low-cost servo arms ship without force sensing, and standard imitation-learning stacks observe joint positions only, so learned policies close the gripper open-loop against contact. For logs that record synchronized joint-position commands and measured positions, the servo tracking error $\delta = a[t-1] - s[t]$ is a contact proxy that can be computed retrospectively, and current stacks discard it. In a pre-registered six-design study at 280 demonstrations on a simulated force-critical task (5 seeds $\times$ 100 paired episodes), every observation that carries this channel explicitly outperforms the channel-free base, while predicting δ as an auxiliary training target does not. The strongest design appends a free-motion-compensated residual, the *lag excess*, reaching 26.6% success against 6.2% for the channel-free base ($t = 4.7$). It is also the only design to beat an action-history baseline at our pre-declared seed resolution (13.8%; 95% CI on the difference $[+2.8, +22.6]$), but it observes one more frame of action history than that baseline, so that comparison varies content as well as form and its information-matched control is absent from this grid. On 546 episodes across 16 public teleoperation datasets, the pre-registered prediction that successful grasps carry higher δ fails: under the frozen outcome proxy, high δ at grasp time is associated with failure, consistent with operators forcing jammed grasps. Code and data at [URL].

Keywords: Imitation Learning, Force Estimation, Low-Cost Manipulation

1 Introduction

Low-cost servo arms and open imitation-learning stacks have made manipulation research broadly accessible: ACT on ALOHA-class hardware [1], diffusion policies [2], and the LeRobot ecosystem [3] with its VLA descendants [4, 5, 6, 7] all train behavior cloning on teleoperated demonstrations, and all share one proprioceptive default: the current measured joint positions, one timestep, nothing else (Sec. 2).

This default fails at contact. The arms carry no force or tactile sensors, so a position-only policy cannot distinguish a seated grasp from an empty close; the failures are crushed objects, unseated grips, and drops during transport. The information needed to distinguish these cases is present in the logs. The servos are proportional position controllers, so when the jaw meets an object the measured position stalls while the command deepens, and the gap reflects the servo’s deflection under load. The authors of ACT state the physics directly: force “is implicitly defined by the difference between” leader and follower positions [1], and a recent intervention study shows on real hardware, across four force-critical tasks, that removing this discrepancy signal from ACT’s action labels breaks them [8]. Yet the stack keeps one operand of that subtraction in the action labels and never provides the difference as an observation; the LeRobot Feetech driver reads only `Present.Position` into the state vector, and the reference ACT implementation raises on `n_obs_steps != 1` [3].

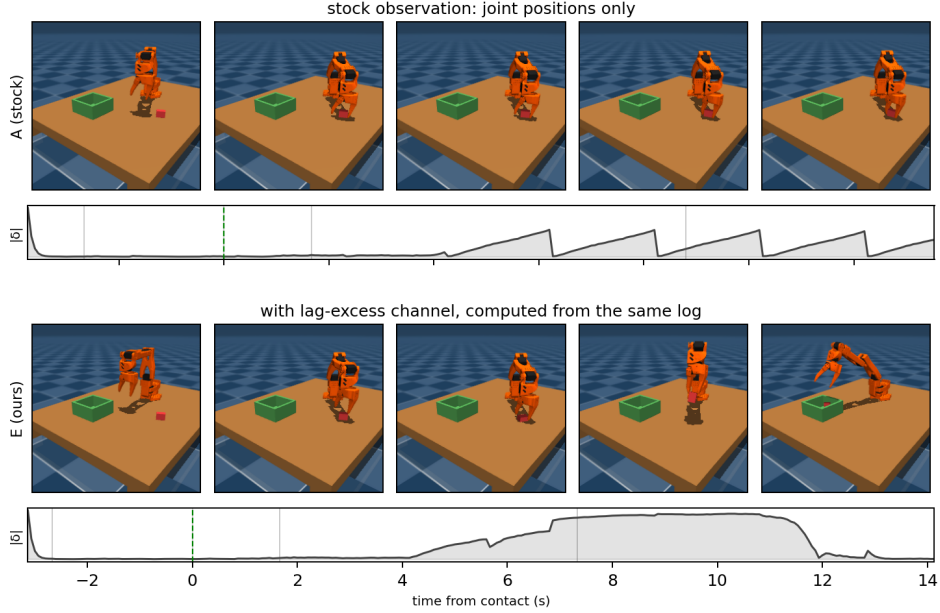


Figure 1: Two policies, identical scene and data budget (280 demonstrations). Top: the stock observation (joint positions only) reaches, grips, and releases without lifting. Bottom: the same network with the contact channel appended to its state vector places the block. The traces plot $|\delta|$ on the jaw joint; the contact plateau in both is the signal the top policy never receives. Both are computed from the same log fields.

37 The channel has a property no current- or torque-based estimator shares: it is recoverable retroactively. Wherever a recorded dataset’s action field holds joint-position commands synchronized with `observation.state` in a common frame, $\delta = a[t-1] - s[t]$ can be computed after the fact; 38
39
40 16 of the 20 public datasets we targeted passed our frozen inclusion gates (Sec. 4.3, which also details the caveats on action semantics). The same signal supports runtime enforcement from bus 41
42 telemetry alone (Sec. 4.4).

43 We ask what a small-data policy must observe to use this channel, and compare six observation 44
45 designs that differ only in which function of the logged pair enters the state vector. The best tested 46
47 form is the *lag excess* $e[t] = \delta[t] - \hat{K}r[t]$, which subtracts a per-joint free-motion following-lag model fit on the jaw-open frames of the training demonstrations, label-free and frozen before training (Sec. 3.1; implementation details in App. A).

48 Three results are established at our pre-declared seed resolution (Sec. 4.2). Channel-bearing obser- 49
50 vation beats channel-free in every input form. The tested auxiliary-prediction baseline recovers none 51
52 of the direct-input gain: the same information moves performance as an input and not as a training 53
54 target. And the lag excess is the only tested design to beat raw action history (26.6% vs. 13.8%, CI 55
56 $[+2.8, +22.6]$); whether its edge over raw δ comes from the subtraction is not resolved at this bud- 57
58 get, so we present it as the recommended instantiation, not a finding about the subtraction itself. On 59
60 the public corpus, a pre-registered prediction fails: high δ at grasp time accompanies failed grasps, 61
62 and the traces attribute this to operators over-commanding at contact (Sec. 4.3). A 40-line guard on 63
64 the same telemetry removes 96% of simulated crush events without retraining (Sec. 4.4).

57 We summarize our contributions below:

- 58 • Characterization of servo tracking error as a contact/load proxy with a validity envelope
- 59 and a resolution-vs-margin design rule (Sec. 3.2, 4.1).
- 60 • A pre-registered six-design observation study at 280 demonstrations: the channel helps as
- 61 an input and not as an auxiliary prediction target (Sec. 4.2).

- A pre-registered 16-dataset corpus study associating high δ at grasp with proxy-defined failure (Sec. 4.3).
- A runtime jaw guard from bus telemetry alone, with paired deleted-success accounting (Sec. 4.4).

2 Related Work

Sensorless force estimation. Momentum-residual observers recover external torque on torque-controlled arms [9, 10]; virtual force sensors extend the idea to low-cost platforms [11], and Hwang et al. [12] estimate torque from commanded and measured positions on hobby servos, prior art for the raw channel — though their estimator is built on a dedicated per-servo dynamic system identification, so while its inference inputs are positions, the model it needs cannot be obtained from an archived log without access to the hardware that produced it; series-elastic actuation made deflection-as-force a design principle [13]. NeuralActuator [14] learns per-actuator models for external-force perception on this servo class, and FACTR 2 [15, 16] feeds estimated external torque into behavior cloning on commodity arms. Our lag excess is a deliberate special case of FACTR 2’s central idea, a linear free-motion baseline fit on positions alone; the restriction makes the signal retroactive, since every method above consumes live current or load telemetry, a calibration stage, or both. Closest to this work, Wong et al. [8] identify the same channel and test it by subtraction: across four real-world force-critical tasks — surface following, insertion, stiffness discrimination, force-based stopping — they remove the leader–follower discrepancy from ACT’s action labels and report severe failures in exactly the force-critical phases, restoring competence with torque proxies drawn from motor current or joint effort. That is the real-hardware form of the observability claim our grid can only make in simulation, and it establishes the core of that claim independently of us. Two differences remain, and they are the ones that matter for a fleet of archived data. Their intervention is subtractive, establishing that the cue matters; ours is additive, asking which function of it a small-data policy should observe. And their remedy needs current or joint-effort telemetry — which the recording stack caps on the contact joint (App. D), and which none of the archived datasets in our corpus records — while the channel studied here needs two position fields every one of those logs already stores.

Observation design in imitation learning. ACT [1], Diffusion Policy [2], OpenVLA [4], π_0 [5], SmolVLA [6], and GR00T N1 [7] all condition on measured joint positions at one or two timesteps; none states a rationale for excluding actuator-error signals. The field’s argument against richer observations is causal confusion: past actions invite copycat shortcuts [17, 18, 19]. Our grid places both effects in one table.

Grasp monitoring and runtime safety. Industrial grippers ship grasp verification from actuator feedback as firmware: Robotiq’s object-detection register fires on commanded-vs-actual discrepancy under a current limit [20], and SCHUNK advertises workpiece-loss detection from an integrated encoder [21]; tactile alternatives require added sensing [22]. Academic safety filters for learned policies [23, 24, 25] consume a dynamics model, a state estimate, or exteroception; the guard of Sec. 4.4 applies the industrial telemetry-only approach to wrap learned policies, and we have not found a filter in the academic literature that runs from servo-bus telemetry alone.

3 Method

3.1 Problem formulation

Let $q[t]$ denote measured joint positions in integer encoder counts and $g[t]$ commanded goal positions, both logged at every control step (in LeRobot format, `observation.state` and `action`). The tracking error and lag excess are

$$\delta[t] = g[t-1] - q[t], \quad e[t] = \delta[t] - \hat{K} \cdot r[t], \quad r[t] = g[t-1] - g[t-2], \quad (1)$$

where $r[t]$ is the one-step command rate and $\hat{K}$ is a per-joint coefficient fit by least squares through the origin on the free-motion (jaw-open) frames of the training demonstrations, computed once and frozen before training (full computation, alignment convention, and normalization in App. A). Each servo runs a proportional position loop, so joint torque obeys $\tau \approx K_p \cdot \delta$ and jaw grip force obeys $F \approx \kappa \cdot \delta_{\text{jaw}}$ for a calibration constant κ in newtons per count. Two conditions make δ meaningful: the logged action must be the command actually applied to the servo, and command-to-state timing must be known and stable. Both hold by construction in our simulated plant; on real logs they are empirical properties of the recording stack (Sec. 4.3). We study which function of the logged pair (q, g) a policy trained on $N = 280$ demonstrations should observe.

3.2 The channel and its envelope

On our simulated plant the jaw calibration is $\kappa = 2.00$ N per count, linear while the jaw is seated and quasi-static, with held-out RMSE 4.3 N over 4–78 N. The same fit carries an intercept of -3.6 N. It is not a measured offset: the fitted forces span 4–78 N, so the smallest δ the fit ever saw is near 3.8 counts and $\delta = 0$ lies outside its domain — the intercept is an extrapolation artifact, which is why the proportional law of Sec. 3.1 carries none. Within the fitted range κ converts *differences* in δ to differences in force; it is used here only in the quantization rule below, which depends on the slope alone, and a reader mapping an absolute δ to newtons with κ and no intercept is off by that constant. These constants describe the modeled plant: the simulator implements the servo loop of Sec. 3.1, so its δ -to-force map is clean by construction; real STS3215 hardware adds stiction, backlash, deadband, saturation, and thermal drift. Bench measurement confirms the saturation and characterizes the baseline registers (App. D); the remainder is still unmeasured. What simulation can characterize is the structure of the proxy: outside the seated quasi-static regime the signal is not a load reading, since free motion contributes following lag (which $e[t]$ removes) and fast transport contributes inertial deflection (App. B, Fig. 5). Encoder quantization yields a design rule: a safe-grip window of force margin M requires an observed quantum on the order of M/κ . We state this as an order and not as a threshold. Treated as a necessary condition it is falsified by our own sweep, which retains most of its success at quanta coarser than M/κ in every margin tested; the rule fires conservatively, ahead of the degradation it predicts, by factors of 1.2–3.1 across three margins (Sec. 4.1). What survives is the mechanism: the failure is a cliff, and its location moves with the margin.

3.3 System, task, and data

All experiments except Sec. 4.3 run on a simulated STS3215-class arm (five joints plus a one-DoF jaw, six servos) with a calibrated contact model. The task is pick-and-place of an object whose retention requires grip force above a lift threshold and whose integrity fails above a crush threshold. A privileged-force oracle passes an admission test on this task and, as a negative control, shows that standard rigid-block pick-and-place cannot distinguish force-aware from force-blind control at all. Demonstrations come from a scripted closed-loop teacher (280 episodes, success-filtered) whose grip regulator acts on the same bus-observable tracking error the arms study, with per-episode grip-depth targets drawn from $U[12, 45]$ counts and injected recovery perturbations; no privileged force sensing enters collection, the teacher’s commands are the applied commands by construction, and the observation arms see only functions of (q, g) . Policies are ACT [1], trained for 12,000 steps at 96 px, identical across arms; at this resolution we cannot rule out a camera deformation cue, but the base arm’s floor performance shows any such cue is insufficient alone. Seed variance was measured before any comparison: $\sigma_{\text{seed}} \approx 10$ points at $n = 100$ evaluation episodes, so 5-seed comparisons resolve gaps of roughly 11 points and we claim nothing finer. Evaluation episodes are common-random-number paired: every arm and seed rolls out the identical episode sequence.

3.4 Observation designs

Six arms share the training recipe and differ only in the proprioceptive observation. **Base (A)**. Measured positions $s[t]$ only: the stock LeRobot observation. **History (B)**. $s[t]$ plus the previous

Table 1: Placement success (%), mean $\pm$ std over 5 seeds (500 paired episodes per arm), with differences against the channel-free base (A) and the action-history baseline (B). Brackets are hierarchical-bootstrap 95% CIs (resampling seeds, then episodes within seeds); an interval containing zero is unresolved at this budget. Best in bold.

Arm	Observation	Success (%)	vs. A	vs. B
A base	$s[t]$	6.2 ± 5.9		
B history	$s + a[t-1]$	13.8 ± 7.7	$+7.6 [-0.8, +16.0]$	
C delta	$s + \delta$	21.8 ± 9.7	$+15.6 [+5.6, +25.2]$	$+8.0 [-2.8, +18.6]$
D aux-target	$s; \delta$ predicted	6.6 ± 4.0	$+0.4 [-6.2, +6.4]$	
E lag excess	$s + e$	26.6 ± 7.8	$+20.4 [+11.6, +29.0]$	$+12.8 [+2.8, +22.6]$
F seat token	$s + \text{seat bit}$	21.8 ± 9.5	$+15.6 [+6.0, +25.2]$	$+8.0 [-2.6, +18.6]$

156 action $a[t-1]$: the minimal change that makes δ recoverable in principle, and the arm the copycat
 157 literature [17, 18] warns about. **Delta (C)**. $s[t]$ plus raw $\delta[t]$: the channel with its motion confound
 158 intact. **Auxiliary target (D)**. $s[t]$ only, with $\delta[t]$ predicted as an auxiliary target through a small head
 159 at loss weight 0.1: the same information as supervision rather than input. **Lag excess (E)**. $s[t]$ plus
 160 $e[t]$: the channel with free-motion lag subtracted. **Seat token (F)**. $s[t]$ plus a latched binary seat bit
 161 from the guard’s frozen stall detector run causally over $(s[t], a[t-1])$: the channel compressed to
 162 one bit.

163 4 Experiments

164 We evaluate: (i) *over what envelope is δ a valid contact/load proxy*; (ii) *which observation design*
 165 *exploits it at small data*; (iii) *does the channel predict grasp outcomes on the public corpus*?

166 4.1 Channel behavior on the simulated plant

167 The calibration of Sec. 3.2 holds in the seated quasi-static regime. Because the simulator implements
 168 the servo loop that generates δ , these numbers validate the proxy against our own plant model; their
 169 value is in mapping where the proxy’s structure breaks. A stall detector on net travel and lag excess
 170 localizes seat events, and constructed adversarial negatives (free-space reversals, transport shakes)
 171 do not fire it within the declared envelope. Coarsening the observed channel eventually collapses
 172 force-aware grasping, and rerunning the sweep at a different margin moves the cliff accordingly —
 173 which is the signature separating a resolution limit from the mere deletion of input bits. The collapse
 174 does not occur at M/κ : at a 100 N limit the channel holds 92% and 75% of its baseline placements
 175 at quanta of 4 and 8, both coarser than the predicted threshold, and the observed cliff runs 1.2–
 176 $3.1 \times$ beyond the prediction across the three margins. The scale constant here is the pre-registered
 177 2.4 N/count of the sweep rather than the $\kappa = 2.00$ of Sec. 3.2; the discrepancy is reported rather than
 178 reconciled, since refitting a constant frozen before the sweep ran would defeat the pre-registration.
 179 TYPICAL_PEAK_N, a single median standing in for a contact peak-grip distribution spanning 19–
 180 56 N, is the likely source, and the rule should be re-derived from that distribution before it is claimed
 181 quantitatively.

182 4.2 Observation design at 280 demonstrations

183 Table 1 and Fig. 3 give the grid: 6 arms $\times$ 5 seeds $\times$ 100 paired evaluation episodes, identical
 184 otherwise. Arm definitions and directional predictions were frozen before implementation, with E
 185 vs. B declared the headline comparison; frozen documents, per-seed outcomes, and all runs are in the
 186 released repository. Alongside Welch’s t across seed means we report 95% CIs from a hierarchical
 187 bootstrap resampling seeds, then episodes.

188 Three results resolve. First, channel-bearing observation beats channel-free in every input form: C,
 189 E, and F each clear the base by at least 15 points with intervals excluding zero (Table 1). Second, the
 190 tested auxiliary-prediction design does not recover the direct-input gain: arm D lands equal to base

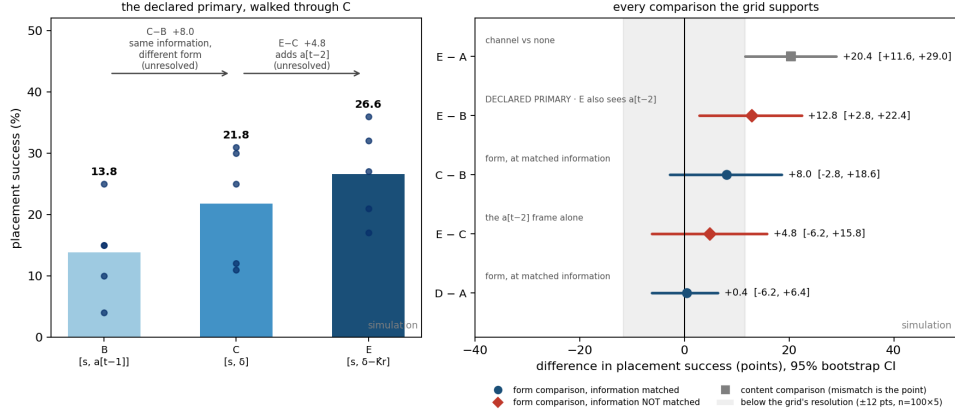


Figure 2: What each comparison in the grid holds fixed. Blue circles vary the *form* of the observation at matched raw content; red diamonds are presented as form comparisons but also differ in content, since the lag excess consumes $a[t-2]$ through the commanded rate and the history baseline does not; the grey square is a content comparison, which is the point of that pair. The shaded band is the below-resolution zone at $\sigma_{\text{seed}} \approx 10$. Intervals are the hierarchical bootstrap of Table 1, resampling seeds and then episodes within seeds.

while the same information as an input triples success; other auxiliary weightings and architectures could in principle close this gap, and none was tuned here. Third, in the declared headline comparison, only the lag-excess form beats action history at resolution, by 12.8 points (Welch's $t = 2.6$ across seed means); raw δ 's 8-point margin over history does not resolve. The remaining pairs stay open, including E against C (difference +4.8, CI [-6.2, +15.8]), so whether E's edge comes from the subtraction or from δ at all is not attributable at this budget; resolving $E-C$ by seed expansion would require roughly 40–60 seeds per arm at this σ_{seed} , and evaluation is already episode-paired. The frozen prediction named E the sharpest arm and expected D, E, and F to clear B; D's share of that prediction failed, as reported above. An interim 3-seed analysis flagged E's margin as resolved and F's as at-the-floor, and the expansion to 5 seeds confirmed both calls, collapsing F's margin over B and preserving E's. Caveats: five seed means per arm, a headline difference of similar magnitude to the declared resolution, and uncorrected secondary comparisons beyond the designated primary one. A best-seed robustness probe under object-property shifts (block mass $\times 0.5$ and $\times 1.5$, friction $\times 0.5$; $n=50$ paired episodes per condition, orderings only) shows the channel arms' margins persist under every shift while the channel-free base returns byte-identical outcomes in all conditions – the signature of a policy that never interacts with the object's dynamics – supporting the reading that channel-bearing policies regulate contact rather than replay a visual sequence (protocol and per-condition table in the repository).

Within those limits the ordering supports one design statement: at small data on this force-critical task, observing the channel is the change that moves performance, and the free-motion-compensated form is the best tested way to observe it.

4.3 Ecosystem study: a pre-registered negative

Does δ predict grasp outcomes corpus-wide? The protocol was frozen before the run: seat detection by the stall detector of Sec. 4.1, δ -at-seat as unsigned tracking error, success proxy defined as the gripper staying closed through transport, and the prediction that successes carry higher δ with Cohen's $d > 0.5$ in at least 10 of 20 datasets. Of the 20 targeted SO-100/SO-101 datasets, 16 passed the frozen inclusion gates with connectivity: 546 episodes, 283 proxy successes.

The prediction fails: 2/16 datasets (pooled within-dataset $d = -0.39$). The channel is informative with the opposite sign: $|d| > 0.5$ in 10/16 datasets, 7/16 significantly inverted (Fig. 4). The traces explain the physics. Teleop leader devices have no hard stop, so operators over-command every

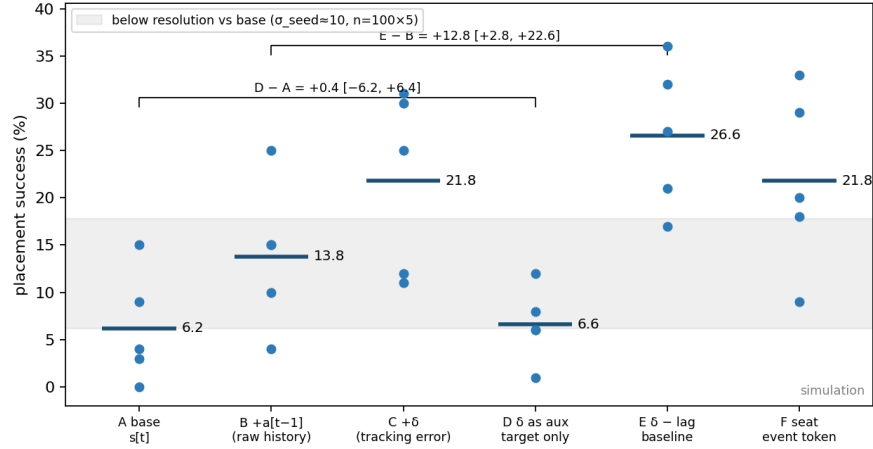


Figure 3: The observation-design grid. Bars are 5-seed means, dots individual seeds (100 episodes each). The gray band spans the below-resolution zone around base given $\sigma_{\text{seed}} \approx 10$. The lag-excess arm (E) is the only design whose margin over the history baseline (B) exceeds seed resolution.

close; the seat flag fires in nearly every episode of 14/16 datasets because empty jaws stall at the mechanical limit exactly as loaded ones do. Successful grasps are stereotyped low- δ closes (in one dataset every clean episode rests at the same 7-count gap), while failures carry 40–80-count frames of an operator forcing a jammed grasp. High δ at grasp time is associated with failure under the frozen proxy. This rules out mining the corpus for success labels through this channel alone, and it supports enforcement: the high- δ events the corpus associates with failure are what the guard caps.

The result also bounds Sec. 4.2’s transfer to retroactive training on real teleop data. When the seat event fires in essentially every episode, a latched seat bit (arm F) carries no information, and raw δ at close time reads commanded depth as much as contact; what stays informative is δ magnitude and its dynamics, which is what the inverted effect sizes measure. Whether the lag-excess form retains its simulated advantage when trained on over-commanded human demonstrations is an open empirical question, not a corollary. Two caveats attach to the corpus numbers: the recorded action is the leader position, which approximates but need not equal the applied follower command, and action-to-state timing is characterized only up to an integer-offset sweep over $k \in \{0, \dots, 5\}$, which leaves these conclusions unchanged (the median per-dataset shift in Cohen’s d across offsets is 0.12) while estimating the physical alignment at three to four recorded frames (App. A).

4.4 Runtime guard

The same telemetry supports enforcement. A 40-line jaw guard (net-travel stall and lag-excess seat detection, a 12-count squeeze budget past seat, a closing-rate limit) runs from bus signals with no dynamics model, no retraining, and no object knowledge; all results are simulated. Paired its detection thresholds make it insensitive to closes slower than its rate gates, so it engages on aggressive closers and not on the gentle scripted teacher (App. C), and on three learned checkpoints over identical paired episodes its first-generation detector (App. C) reduced crush events from 85 to 3 of 180 (96%) across the crush-capable tiers, with median peak force over those same episodes dropping from 33–92 N to 13–26 N. It also removes every success in the experiment: 12 of 270 unguarded become 0 of 270, and no guarded cell scores above zero. The policies are weak to begin with here (4.4% unguarded, and the channel-free arm scores zero either way), so this is the guard deleting the few successes that existed rather than dismantling a working policy — but a wrapper that never succeeds is not yet a deployable one. Demonstrations must record applied commands, and wrapped controllers must be closed-loop on contact. Details, figure, and integration accounting in App. C. On hardware the register this guard would otherwise read is capped by the recording stack itself and saturates through contact, while the channel does not (App. D).

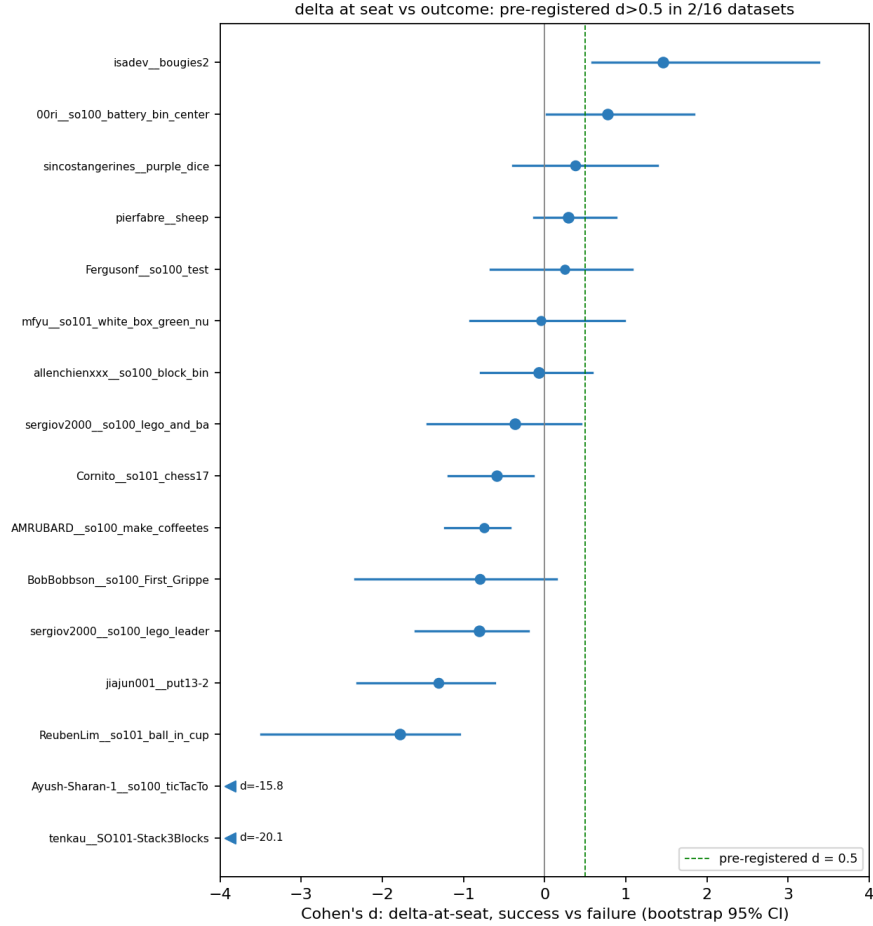


Figure 4: Per-dataset Cohen’s d of δ -at-seat between proxy successes and failures, bootstrap 95% CIs, for the 16 datasets passing the frozen gates (attrition documented in the repository). Positive d was the pre-registered direction. Two datasets exceed $d = 0.5$ as predicted; six are significantly inverted; two task-repetitive datasets reach $d \approx -16$ and -20 because their stereotyped successes have near-zero δ variance.

253 5 Limitations

254 All learning and guard results are simulation, on one gripper and one object family; every quantity
 255 in newtons inherits the simulator’s servo and contact model until bench calibration lands (App. D).
 256 The task is constructed to require force-band control, so the grid measures observation design on
 257 a force-critical task, not manipulation in general. The lag-excess model is a first-order correction;
 258 an offline ladder of richer free-motion predictors (intercept, direction and acceleration terms, short
 259 causal FIR) improves held-out free-motion residual by at most 1.9% over the scalar model on the
 260 jaw, so the linear choice is not leaving obvious compensation on the table, though learned predictors
 261 in the style of Oh et al. [15] remain untested as policy inputs. The seed budget leaves $C-B$ and
 262 $E-C$ open, so the value of the subtraction itself is not established.

263 The most consequential open question concerns operating point. The grid runs at 280 demonstra-
 264 tions, 96 px and 12k steps. At a larger budget — 600 demonstrations, 224 px, 50k steps, where
 265 absolute success reaches 57–63% rather than the 6–27% reported here — a two-seed comparison of
 266 the raw channel against the action-history baseline is indistinguishable (60% vs. 57%, with a within-
 267 policy seed spread of 12 points, four times the between-policy gap). That cell tested C against B ; E ,
 268 the design this paper recommends, has not been run at that budget. Two readings are therefore live

and we cannot currently separate them: that the channel’s advantage is a small-data effect which disappears where policies become useful, or that the free-motion subtraction is precisely what survives scaling and the raw form is not. What the small-data ladder does support independently of either is that policies with no action-history input in any form plateau at 16–24% lift commitment across three data recipes while channel-bearing policies reach 47–76%: the information appears necessary, and it is the *form* whose value at scale is unresolved. This pattern is not peculiar to our setup: Zeng et al. [26] sweep context length against dataset size and find the ordering between short- and long-context policies reverses between 200 and 1000 demonstrations, with dataset size disproportionately favouring the longer-context policy. Our 280-to-600 window sits inside that crossover, so the tie we observe is the expected behaviour of a known effect rather than an unexplained inconsistency — which also means a single budget cannot identify the mechanism, and any arm we compare must be compared at more than one.

A second caveat bears on the primary comparison specifically. The lag excess $e[t] = \delta[t] - \hat{K}r[t]$ depends on $a[t-2]$ through the commanded rate, while the action-history baseline B observes $s[t]$ and $a[t-1]$ only, and $\hat{K}$ is far from zero (0.74–0.84 across joints, 0.80 on the jaw). E therefore consumes one more frame of action history than the baseline it is compared against, and $E-B$ confounds the free-motion subtraction with that additional frame. The comparison that isolates the subtraction holds the raw inputs fixed — E against a two-step raw history $[s[t], a[t-1], a[t-2]]$ — and that arm is not in the present grid. $C-B$ is by contrast a clean representation test, since C and B observe the same raw quantities — and it does not resolve at our seed budget ($+8.0$, $t = 1.44$). The decomposition is exact: $E-B = (E-C) + (C-B) = +4.8$ ($t=0.86$) $+ 8.0$ ($t=1.44$) $= +12.8$, so the one pair clearing our resolution bar is significant only as the sum of two components neither of which is separately claimable at $n = 5$ seeds. A wider audit of the arms’ information dependence sharpens this: of the design comparisons the grid draws conclusions from, only $C-B$ and $D-A$ hold raw content fixed while varying form. F in particular consumes an entire episode prefix, because its seat detector latches once and never resets, yet does not beat E — so a simple information ordering does not explain the grid either. $D-A$ is matched by construction, the channel entering only as an auxiliary target, which makes the observe-versus-predict null the grid’s most securely grounded design result. Nothing here bears on the channel-versus-channel-free result ($E-A = +20.4$, $t = 4.7$); what is unresolved is the narrower claim that the lag excess specifically improves on action history. Running E and the matched two-step baseline at both budgets is the experiment we would prioritize. On real logs, δ depends on action semantics and timing that the corpus study characterizes only up to an integer-offset sweep. The corpus success proxy is unvalidated against ground truth, and multi-grasp tasks were excluded by design. The grid is measured on one demonstration collection; a three-seed recollection (App. C) surfaces collection-level resampling variance that our comparisons do not yet average over. The channel is valid only seated and quasi-static.

6 Conclusion

Position-command logs contain a retrospectively computable contact proxy: the servo tracking error, valid within a quasi-static envelope our simulated plant makes measurable. At 280 demonstrations, policies must observe this channel; predicting it as the tested auxiliary target recovers none of the gain, and the free-motion-compensated form is the best tested instantiation. Across 16 public datasets high δ at grasp time accompanies failure, which rules out free outcome labels from this channel alone and supports its use for enforcement. What remains open requires hardware: bench calibration, retroactive training on real teleop logs, and a copycat-style probe of when history helps versus hurts in behavior cloning.

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382 A Channel computation and implementation details

383 All signals are in integer encoder counts at the recorded cadence (stride-2 frames of a 30 Hz bus,
 384 i.e. 15 Hz): $\delta_j[t] = a_j[t-1] - s_j[t]$, where a is the commanded and s the measured position, the
 385 $k=1$ alignment convention. An offset sweep over $k \in \{0, \dots, 5\}$ on the public corpus leaves
 386 every Sec. 4.3 conclusion unchanged (the median per-dataset shift in Cohen’s d across offsets is
 387 0.12) while estimating the physical alignment at three to four recorded frames; the bench latency
 388 measurement (App. D) will pin it. The commanded rate is the first difference of commands, $r_j[t] =$
 389 $a_j[t-1] - a_j[t-2]$, zero at the first two frames of an episode. The lag model is a single per-joint
 390 scalar through the origin, $\hat{K}_j = \sum r\delta / \sum r^2$, fit over frames whose jaw command exceeds its 30th
 391 percentile; the jaw is the task’s only contact, so jaw-open frames are free motion. Fitted values
 392 span $\hat{K} = 0.74\text{--}0.84$ across joints on our training set (and move by at most 0.09 when the free-
 393 motion percentile criterion is varied from the 10th to the 50th), computed once on the training
 394 demonstrations and frozen. The policy input is the concatenation $[s_{\text{norm}}, (\delta - \hat{K}r - \mu_\delta)/\sigma_\delta]$; joint
 395 states and actions are standardized by dataset statistics, while δ , e , and the aux target use a fixed scale
 396 of $\sigma_\delta = 8$ counts with $\mu_\delta = 0$, so no channel statistic leaks from evaluation. The same statistics
 397 are saved beside the checkpoint and applied identically at evaluation, where $a[t-1]$ is the previously
 398 issued command, so the feature is computable from exactly what the deployed stack already has.
 399 Appended channels concatenate onto the state vector, leaving the network otherwise unchanged.

400 B Mechanism on a single grasp

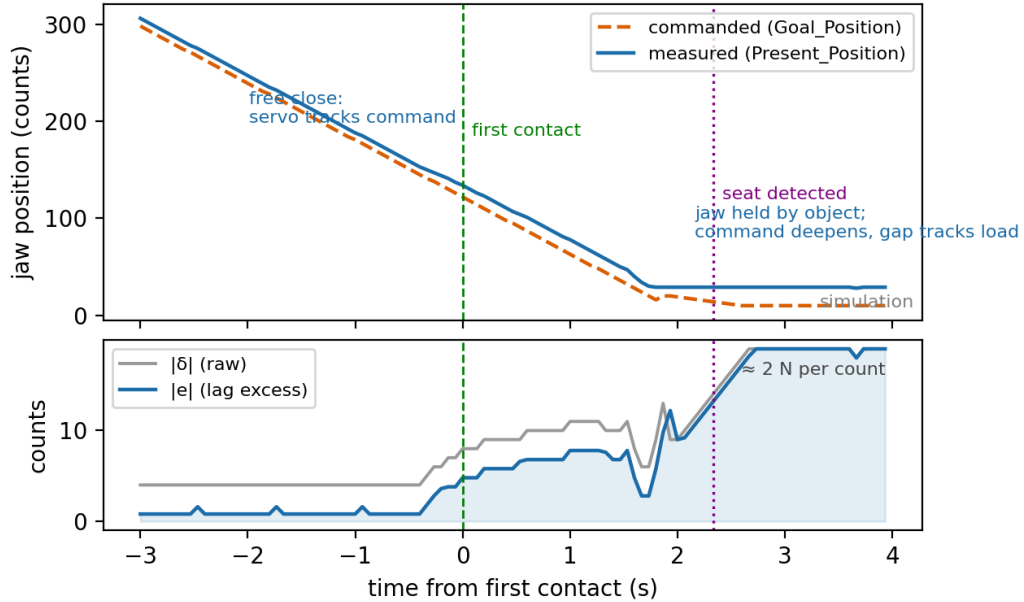


Figure 5: The channel on one servo and one grasp. The commanded position (dashed) and measured position (solid) track closely through the free close, separated only by the small following lag. At first contact (marked) the measured position stalls while the command continues to deepen; the growing gap is the load reading, $\kappa \approx 2$ N per count on the simulated plant. The detector’s seat event (marked) fires when net travel stalls while lag excess grows. The lower panel plots $|\delta|$ and the lag excess $|e|$ over the same frames; $|e|$ sits near zero through the free close by construction and inherits the load reading after contact.

C Guard details

The guard combines net-travel stall and lag-excess seat detection, a squeeze budget of 12 counts ($\approx +24$ N on the simulated plant) past seat, and a closing-rate limit. Two regimes follow from the detector’s thresholds. On aggressive closers it binds: learned policies commanding fast or deep closes are capped near 40 N against raw peaks of 99–127 N, and an open-loop clamp reference commanding 360 N grips is held to 10–28 N. On the gentle closed-loop teacher (~ 0.3 counts per frame in its fine pass, below both detection gates) it never fires: a paired run shows 25/30 placements wrapped versus 28/30 raw at statistically indistinguishable ~ 39 N grips, i.e. no enforcement occurred and the run measures wiring overhead only. Whether threshold-level insensitivity to gentle closes is acceptable conservatism or a coverage gap depends on the deployment’s hazard model; we state it as a measured property of these thresholds.

An independently recollected demonstration set (same teacher, recipe, and budget; fresh episodes) trained at three seeds per arm neither resolves any per-arm difference from the original grid nor reproduces its point-estimate ordering at that reduced seed count (the lag-excess arm’s fresh-collection seeds span 7–27 against the original 17–36, Welch’s $t = 1.6$). Across the six arms the three channel-bearing designs’ point estimates fell while the three channel-free designs’ rose, none individually resolving; four marginal statistics of the two collections (tracking-error distributions, command floors, command-rate distributions, per-episode grip-depth diversity) are indistinguishable, so we treat this as unattributed collection-level resampling variance – a component above seed variance that the headline comparison of Sec. 4.2 does not yet average over, and that hardening it requires spanning with seed expansion over multiple collections. The crush-elimination A/B predates the detector’s final revision: on three learned checkpoints over identical paired episodes the first-generation detector (single-step stall test, 4-count budget) reduced crush events from 85 to 3 of 180 across the two crush-capable tiers (Fig. 6), with median peak force over those same 180 episodes dropping from 33–92 N to 13–26 N; across all nine cells, including the no-crush tier where crushing is impossible, the unguarded range is 33–131 N. A rerun under the final detector is pending alongside hardware validation. The guard also deletes successes, and it deletes all of them: unguarded totals are 12 successes in 270 episodes, guarded totals are 0 in 270, and every one of the nine cells scores 0/30 guarded. The losses fall on two arms — the quantized arm gives up 7/30 at a 131 N median and 1/30 at the 120 N tier, the delta arm 2/30 and 2/30 — while the channel-free arm scores zero with and without the guard. Since the unguarded rate is 4.4%, this is a guard removing the few successes that existed rather than a working policy being dismantled, but it means the enforcement layer as tested trades all measured task success for a 96% crush reduction. Integration requires two properties: demonstrations must record applied commands rather than intentions, and any wrapped controller must be closed-loop on contact, since the rate limiter breaks open-loop grasp timing by construction. Collecting demonstrations under *active* guard enforcement would require a detector variant sensitive to gentle closes and remains untested.

D Hardware: the reviewer’s baseline on a physical STS3215

The first question this paper invites is why estimate contact from tracking error when the servo already reports `Present_Load` and `Present_Current` over the same bus. This appendix answers that question on hardware and states plainly what it does not answer: the newton scale is not yet measured, so every force quantity in the body still inherits the simulator’s contact model.

D.1 Static: the register is the channel, rescaled

Protocol. A payload staircase on one STS3215 in an SO-101 shoulder-lift joint. Twelve hanging masses spanning 63.7–319 g, each held at five arm poses, logging `Present_Load` (address 60), `Present_Current` (69), `Present_Position` (56) and `Goal_Position` (42) at static equilibrium; $\delta = \text{Goal} - \text{Present}$, in encoder counts. Masses are known from package labelling rather than weighed, so their systematic error is unquantified. Three masses were measured twice (106, 149,

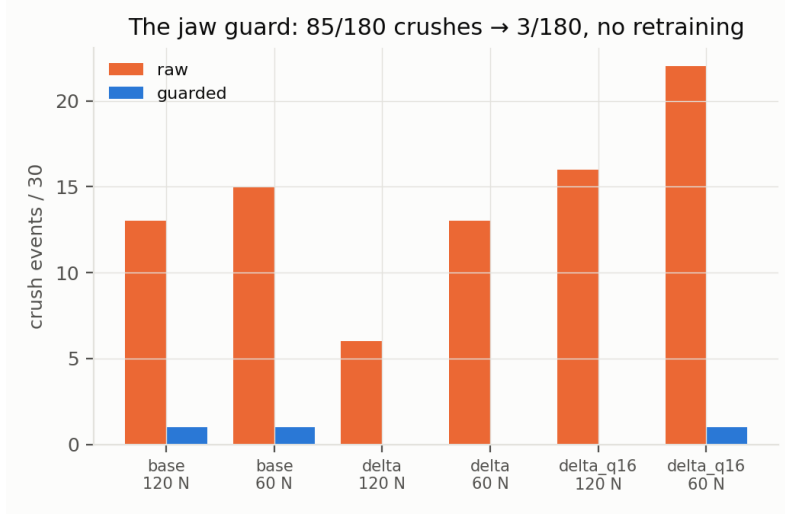


Figure 6: Crush events per 30 paired episodes for three learned checkpoints (base, delta, and a 16-count-quantized delta arm) at two crush tiers, raw (orange) versus guarded (blue), under the first-generation detector. Identical checkpoints and episode seeds; the guard acts only through bus telemetry at runtime. Totals: 85 of 180 raw, 3 of 180 guarded.

449 191 g), and those replicates are used below as the repeatability estimate; they bound the random
 450 component only.

451 **Poses are fit separately.** Pooling them regresses across five different lever arms at once and inflates
 452 every residual. All intervals below are 95% bootstrap percentiles over 10,000 resamples of the
 453 twelve masses.

pose	δ slope (counts/g)	R^2_δ	R^2_{load}	R^2_{curr}	replicate $ \Delta_\delta $
3	-0.0194 ± 0.0019	0.910 [0.85, 0.98]	0.910	0.014	0.33
1	-0.0323 ± 0.0044	0.845 [0.72, 0.95]	0.845	0.726	1.33
2	-0.0435 ± 0.0145	0.473 [0.10, 0.81]	0.473	0.503	7.00
0	-0.0236 ± 0.0126	0.261 [0.01, 0.71]	0.279	0.041	4.00
4	-0.0115 ± 0.0119	0.085 [0.00, 0.63]	0.077	0.034	2.67

Table 2: Per-pose fits against true hanging mass, ordered by R^2_δ . Replicate $|\Delta_\delta|$ is the mean absolute difference in δ between the two independent measurements at each repeated mass, in counts.

454 **The register carries the same information as the channel.** Pooled over all sixty measurements,
 455 $\text{Present_Load} = -4.70 (\pm 0.10) \delta + 0.1$ with $R^2 = 0.976$ [0.969, 0.982] (Fig. ??, right). Per pose
 456 the two track the true mass to within 0.017 of each other in R^2 , agreeing to three decimals in three
 457 of five poses. They are one signal on this joint, so the register cannot be a better instrument than the
 458 channel — only a differently scaled one. This is the claim the appendix is for, and it is the one the
 459 data supports most strongly.

460 **What limits it is repeatability, not lever arm.** The spread in Table 2 is large, and it does not follow
 461 slope magnitude: pose 2 has the steepest response to mass and still resolves poorly, while pose 3 has
 462 a middling slope and the tightest fit. It follows the replicate column instead. Dividing repeatability
 463 by slope gives the mass a single measurement can distinguish: about 17 g at pose 3 and 41 g at pose
 464 1, against 160–230 g at the three poorly repeating poses. The channel’s usable resolution on this
 465 joint is therefore a property of where the arm is standing, not a single number, and the favourable
 466 figure is the exception rather than the summary.

467 **Current is not established as the better instrument, or the worse one.** δ attains the higher R^2
 468 in four of five poses, which under an exact paired sign test is $p = 0.375$ — at $n = 5$ poses that is

uninformative in either direction, four coin flips landing the same way. We report the ordering and claim nothing from it.

D.2 Dynamic: the baseline loses its range exactly at contact

Fifty teleoperated demonstrations on the physical arm pair (21,476 frames, a pick-and-place with a single overhead camera and no wrist view) show what the static staircase cannot. `Present_Load` pins at 500 in 16.5% of frames, and all fifty episodes reach it. The saturation is not spread evenly: it coincides with contact, where $|\delta|$ averages 12.35° against 2.09° elsewhere. Inside those frames the register reports a single constant value while δ still spans 8.01 – 31.01° , twenty-three degrees of range the baseline cannot express. Where the register is not clipped the two agree as the static data predicts ($r = -0.853$ unsaturated, -0.922 overall).

D.3 The ceiling belongs to the stack, not to this arm

Reading the torque-limit register (address 48) returns 1000 for joints one through five and 500 for the gripper alone. `Present_Load` is full-scale 1000, so 500 is a configured cap on the one joint that makes contact. It is not a local misconfiguration: LeRobot [3] writes it for every SO-100/SO-101 follower it connects, inside the configuration path that connection always invokes, together with a matching halving of the current-protection register — both justified, reasonably, as burnout protection.

Both baselines are therefore capped, on the contact joint, by the library the fleet records through. δ requires no register and no limit; it is arithmetic over two numbers every one of those logs already stores, and is untouched by the same configuration.

The register read is from one arm, and we state the limit rather than leave it inferred: a single follower reading 500 on its gripper is equally consistent with that unit having been flashed that way. The load-bearing evidence is therefore the code path, which writes the value unconditionally on every connection, not the measurement, which confirms the value is present on the one unit we have. The discriminating test is cheap and we name it so a reader can run it: the leader configuration path performs no torque-limit write at all, so a leader gripper reading full scale against a follower gripper reading 500, on arms of the same model and flash, would establish that the stack writes the cap rather than the hardware shipping with it. Until that is done the claim is that the library sets this ceiling on every follower it connects, which the source establishes, rather than that we have observed it across a fleet.

D.4 What is still not measured

Four gaps, stated because they bound the claims above.

No load cell has been applied, so the newton scale is unmeasured and the simulator’s 2.0 N/count remains the provenance of every force figure in the body. Hanging masses give torque in the gravity direction on one joint; they do not give newtons per count in the jaw’s closing direction, which is the quantity the task actually depends on. The static staircase is one joint, leaving stiffness measured at $n = 1$ and its per-servo-versus-per-joint prediction untested. It is static only: no three-speed sweep, no opening and closing branches for hysteresis, no drift checks, and no command-to-state latency distribution — the last of which Sec. 3.1’s alignment argument still resolves by corpus sweep rather than by measurement. Finally, the current half of the saturation result is inferred from the configuration write rather than observed, since these recordings log position and load but not current.

Comparison rows for a completed calibration remain the simulator’s 2.0 N/count, Hwang et al. [12], and NeuralActuator [14].